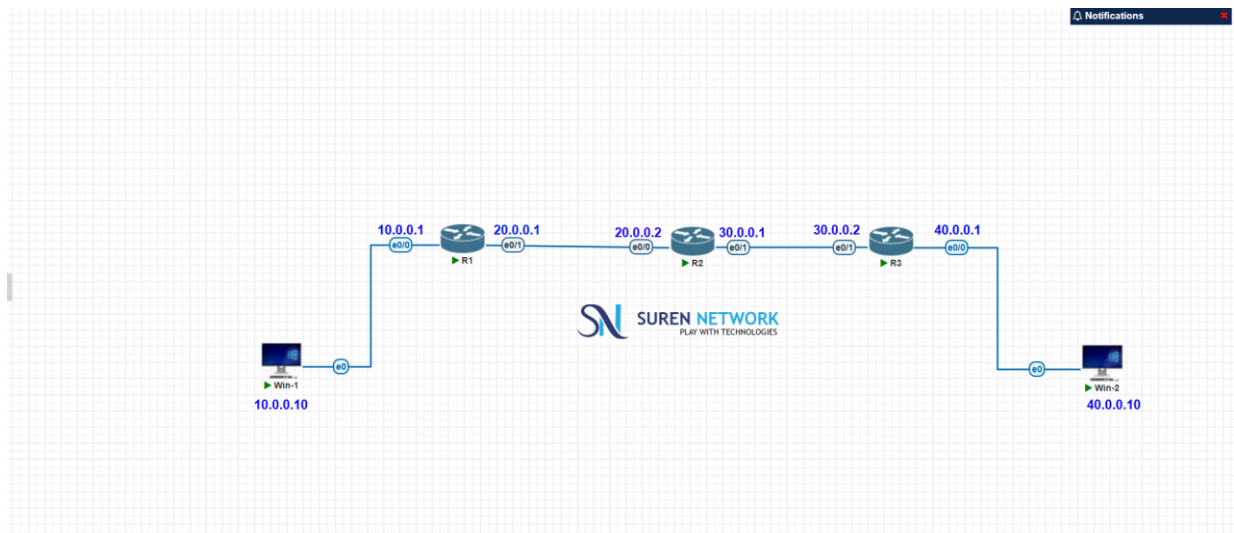


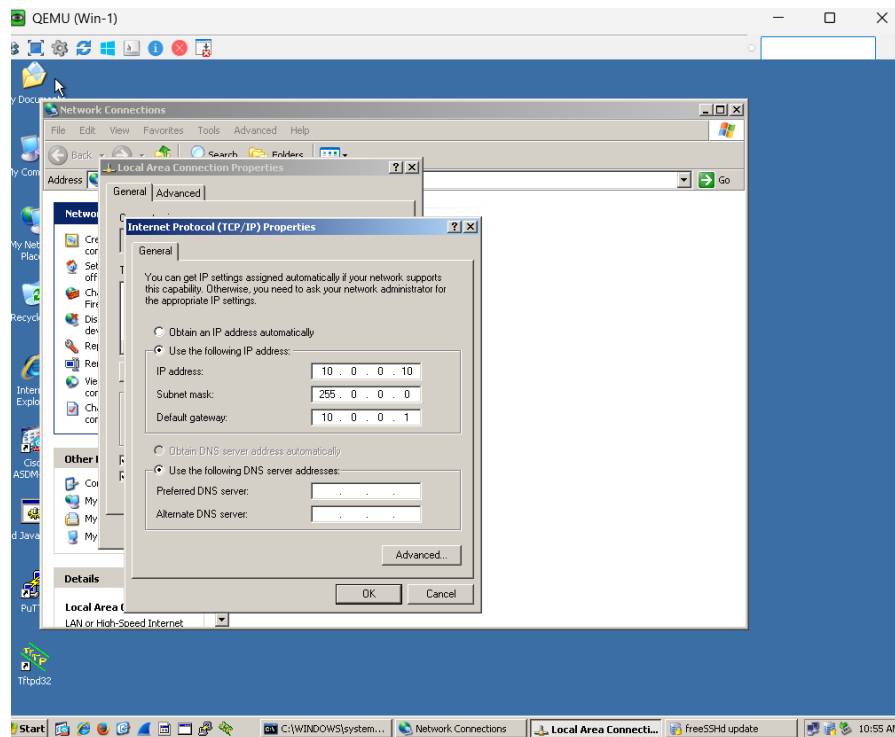
Communication Between Two Windows PCs Through Multiple Routers:

To configure and verify communication between two Windows PCs located on different networks using three routers.



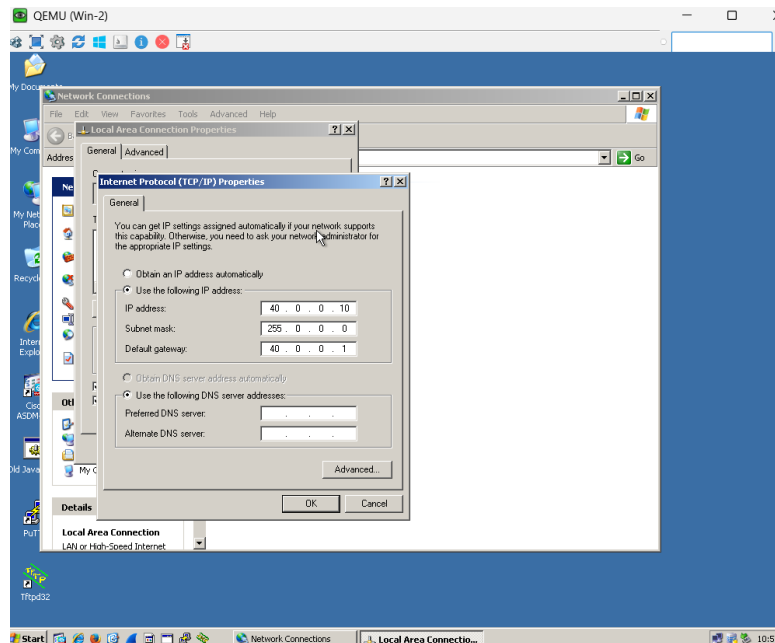
Win-1

Configure:



Win-2

Configure:



Router Configuration

R1 Configuration:

```
enable
```

```
configure terminal
```

```
interface ethernet 0/0  
ip address 10.0.0.1 255.0.0.0  
no shutdown  
exit
```

```
interface ethernet 0/1  
ip address 20.0.0.1 255.0.0.0  
no shutdown  
exit
```

Static Route on R1:

```
ip route 40.0.0.0 255.0.0.0 20.0.0.2
```

```
ip route 30.0.0.0 255.0.0.0 20.0.0.2
```

```
R1
R1#sh
Aug 10 09:34:25.364: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      10.0.0.0/8 is directly connected, Ethernet0/0
      10.0.0.1/32 is directly connected, Ethernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      20.0.0.0/8 is directly connected, Ethernet0/1
      20.0.0.1/32 is directly connected, Ethernet0/1
    30.0.0.0/8 [1/0] via 20.0.0.2
    40.0.0.0/8 [1/0] via 20.0.0.2
R1#
```

R2 Configuration:

```
Enable
configure terminal
interface ethernet 0/0
ip address 20.0.0.2 255.0.0.0
no shutdown
exit
interface ethernet 0/1
ip address 30.0.0.1 255.0.0.0
no shutdown
exit
```

Static Routes on R2

Route to the Win-1 network:

```
ip route 10.0.0.0 255.0.0.0 20.0.0.1
```

Route to the Win-2 network:

```
ip route 40.0.0.0 255.0.0.0 30.0.0.2
```

```

R2#s
*Aug 10 09:36:08.567: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      20.0.0.0/8 is directly connected, Ethernet0/0
      20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      30.0.0.0/8 is directly connected, Ethernet0/1
      30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2#

```

R3 Configuration:

```

enable
configure terminal

interface ethernet 0/1
ip address 30.0.0.2 255.0.0.0
no shutdown
exit

interface ethernet 0/0
ip address 40.0.0.1 255.0.0.0
no shutdown
exit

```

Static Route on R3

R3 needs a route to the **10.0.0.0/24** network.

```

ip route 10.0.0.0 255.0.0.0 30.0.0.1
ip route 20.0.0.0 255.0.0.0 30.0.0.1

```

```

R3#show
*Aug 10 09:37:37.498: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      30.0.0.0/8 is directly connected, Ethernet0/1
      30.0.0.2/32 is directly connected, Ethernet0/1
C    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      40.0.0.0/8 is directly connected, Ethernet0/0
      40.0.0.1/32 is directly connected, Ethernet0/0
R3#

```

Successfully verified end-to-end connectivity between Win-1 and Win-2 using the `ping` command. Both systems communicated successfully with 0% packet loss, confirming proper routing and network connectivity.

